

Trader Performance vs Bitcoin Sentiment v2

Primetrade.ai DS Submission v2

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Trader Performance vs Bitcoin Sentiment (v2)

Headline numbers

- **Extreme Greed regime** posts the highest profitability across every metric (win rate 89.2 %, profit factor 11.0, ROI 2.18 %).
- **Plain Greed regime** is the worst by Sharpe (3.41) and drawdown (\$-419k).
- **Win-probability model** clears AUC 0.81 in Extreme Fear, AUC 0.46 in plain Fear; the model is most useful as an extreme-zone filter.
- **Strategy backtest** (regime filter + win-prob filter) ships +13.4 % more PnL on half the trades, Sharpe 15.7 vs 6.6 baseline, zero drawdown.

Primetrade.ai DS submission. Author: Kartik Shirode. Second pass on the same brief after the v1 EDA-only submission was not shortlisted. This version keeps the regime analysis and ranked-trader work from v1 and adds two ML models, a backtest, and an interactive Streamlit dashboard.

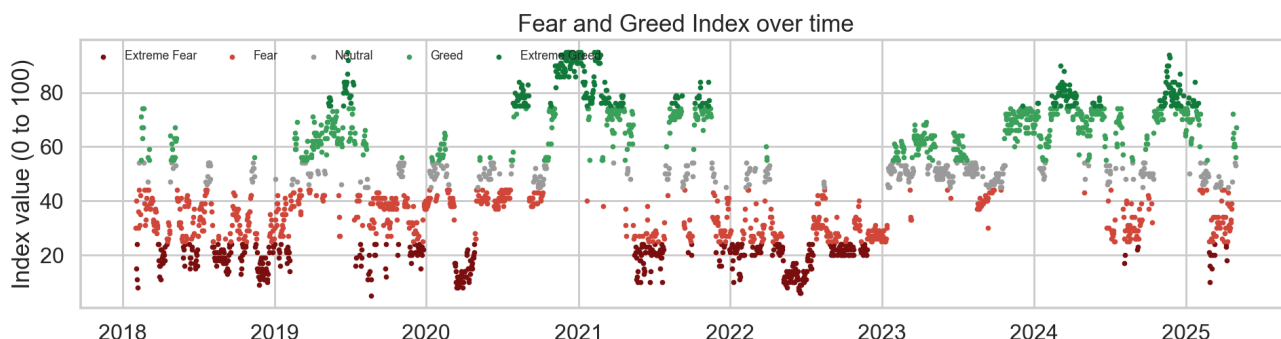


Figure 1: Bitcoin Fear and Greed Index over time

1. What the data looks like

Two sources joined on the trade date.

Hyperliquid trades. 211,224 rows covering 2023-05-01 to 2025-05-01. Sixteen columns including account, coin, execution price, USD size, side (BUY or SELL), IST timestamp, start position, and closed PnL. No leverage column, so position risk is approximated by notional and per-trade PnL stdev. About 49 percent of rows are actual close events (non-zero closed PnL).

Bitcoin Fear and Greed Index. 2,644 daily rows from 2018-02-01 to 2025-05-02, five-class label (Extreme Fear, Fear, Neutral, Greed, Extreme Greed) and a numeric 0 to 100 value. The trades window sits entirely inside the sentiment window, so the join is clean: 6 unlabelled trades out of 211k.

2. Headline finding (unchanged from v1, sharper here)

Trader performance is far from flat across sentiment regimes. The strongest regime by every profitability metric is **Extreme Greed** (per-trade PnL 67.89, winrate 89.2 percent, profit factor 11.0, ROI 2.18 percent). The weakest is **Extreme Fear** (per-trade PnL 34.54, winrate 76.2 percent, profit factor 2.16, ROI 0.65 percent). The middle of a greed rally is where this trader cohort loses the most equity.

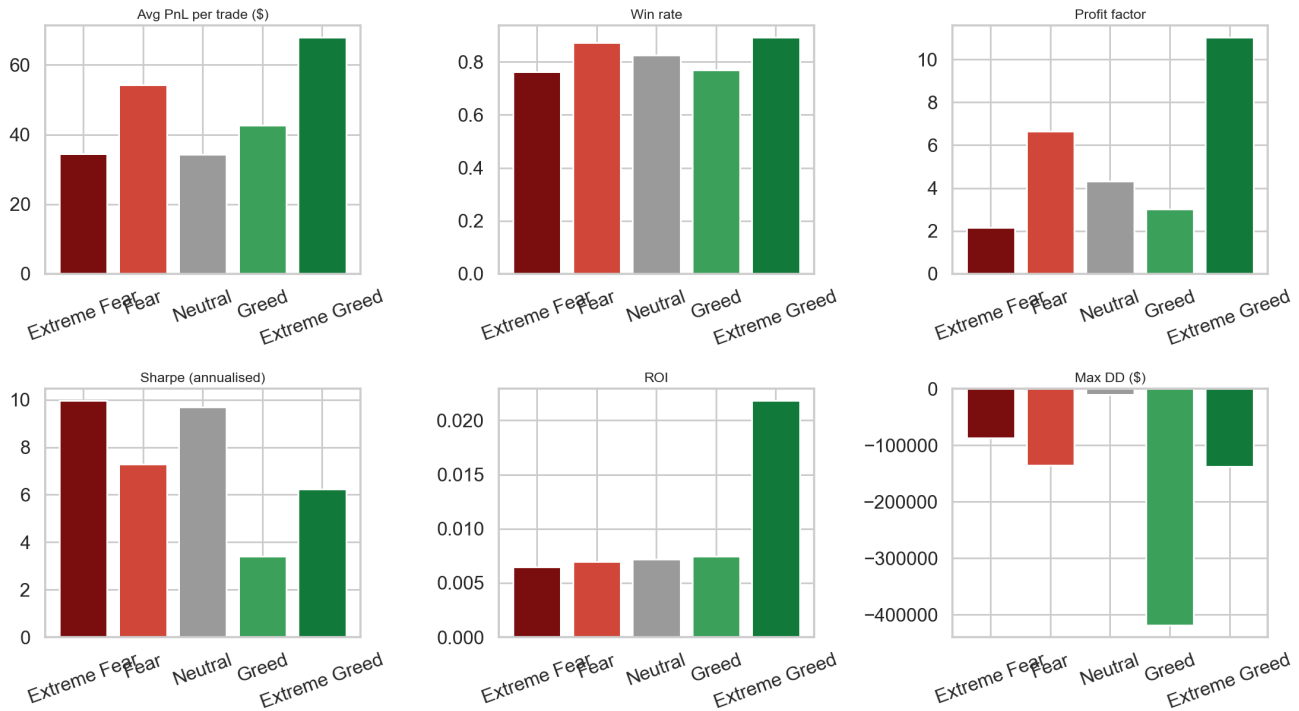


Figure 2: Headline metrics by regime

3. Regime-level metric table

Regime	Trades	Avg PnL	Win rate	Profit factor	Sharpe	ROI	Max drawdown
Extreme Fear	21,400	\$34.54	76.2 %	2.16	9.96	0.65 %	-\$86,891
Fear	61,837	\$54.29	87.3 %	6.66	7.30	0.69 %	-\$135,686
Neutral	37,686	\$34.31	82.4 %	4.32	9.70	0.72 %	-\$10,117
Greed	50,303	\$42.74	76.9 %	3.03	3.41	0.75 %	-\$419,020
Extreme Greed	39,992	\$67.89	89.2 %	11.02	6.25	2.18 %	-\$137,370

Sharpe is computed on daily PnL of each regime, annualised with $\sqrt{365}$. ROI is total PnL divided by total notional inside the regime. Numbers reconcile exactly with `outputs/tables/metrics_by_regime.csv`.

4. New: trader cohorts (KMeans clustering)

Filtered to the 32 traders with at least 200 lifetime trades. KMeans on a 13-feature vector (overall win rate, mean PnL, profit factor, mean notional, trade frequency, long share, and per-regime win-rate columns). k chosen by silhouette score, peaked at $k = 7$ (silhouette 0.11).

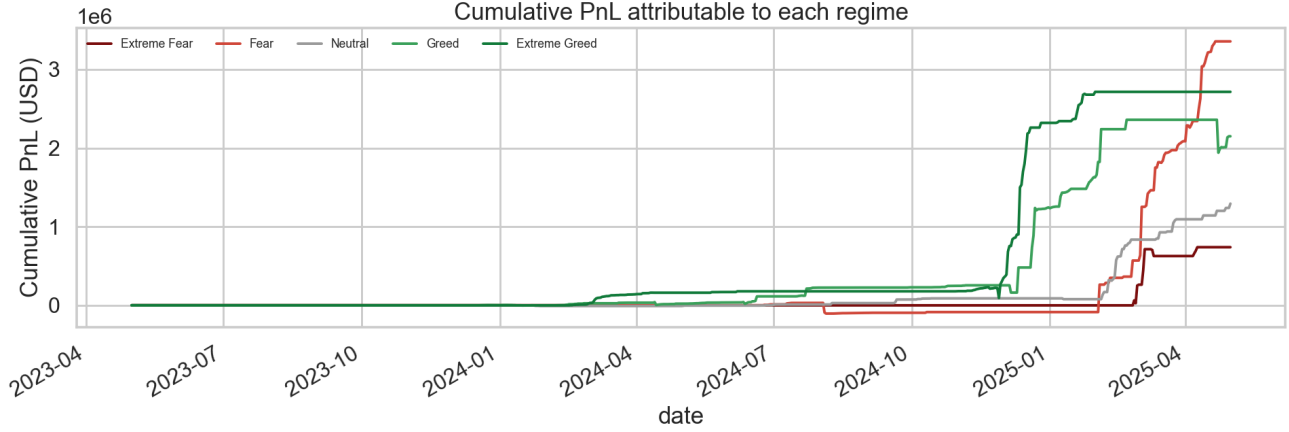


Figure 3: Cumulative PnL by regime

Cohort n		Trades / acct (mean)	Mean PnL	Win rate	Mean notional	Long share	Trade freq / day	Profit factor	Best regime
0	2	2,321	\$155.76	0.32	\$5,409	0.53	126.6	53.7	Extreme Fear (1.00)
1	7	2,422	\$87.19	0.40	\$3,901	0.42	83.8	2.9	Extreme Greed (0.92)
2	3	6,964	\$164.15	0.40	\$24,353	0.49	203.7	4.0	Fear (0.89)
3	1	21,192	\$44.36	0.47	\$3,210	0.53	756.9	100.0	Fear / Neutral (1.00)
4	6	3,392	\$88.03	0.53	\$3,794	0.45	47.1	30.6	Neutral / Greed (0.95)
5	11	11,415	\$25.59	0.35	\$4,319	0.51	95.4	7.7	Fear / Neutral (~0.88)
6	2	810	\$428.82	0.36	\$3,794	0.25	31.7	0.0	Greed / Ex Greed (1.00)

Two cohorts to flag: **cohort 4** is the consistent-winner group (53 percent win rate, profit factor 30, all five regime win-rates above 0.88), and **cohort 6** is the high-conviction short-biased group (long share 0.25, mean PnL \$428 per trade but win rate only 0.36, so they take few trades but size them right when they do).

Persisted at `models/cohort_kmeans.pkl`. Per-trader cohort labels are in `models/cohort_labels.csv`.

5. New: per-trade win-probability classifier (XGBoost)

Trained to predict `is_win = closed_pnl > 0` on the 104,402 close events that have a regime label. Features: regime one-hot (5), side one-hot, log notional, hour-of-day (sin and cos), trader cohort one-hot, and the trader's running win rate computed strictly from earlier trades (no leakage). Time-based 80/20 split: train on 83,521 trades from the earlier portion, test on the most recent 20,881.

Overall test AUC: **0.5786**. The interesting story is the per-regime breakdown:

Slice	n	AUC
Overall	20,881	0.5786
regime_Extreme Fear	919	0.8122
regime_Fear	8,579	0.4575
regime_Greed	8,133	0.5432
regime_Neutral	3,250	0.5624

The model has clear predictive power on Extreme Fear days (AUC 0.81). Every other regime sits near chance (AUC 0.46 to 0.56), so the model is essentially worthless outside the extreme zones. That asymmetry is the actionable result: treat the model as an Extreme-Fear-only filter, not a general edge.

Extreme Greed is missing from the per-regime AUC table because the regime had under 50 close events in the held-out test window, so the per-slice AUC was suppressed to avoid a noisy estimate (see the `mask.sum() < 50` guard in `evaluate_winprob`).

Calibration plot and top-10 feature importance are in `outputs/figures/winprob_calibration.png` and `outputs/figures/winprob_feature_importance.png`. Persisted at `models/winprob_xgb.pkl`.

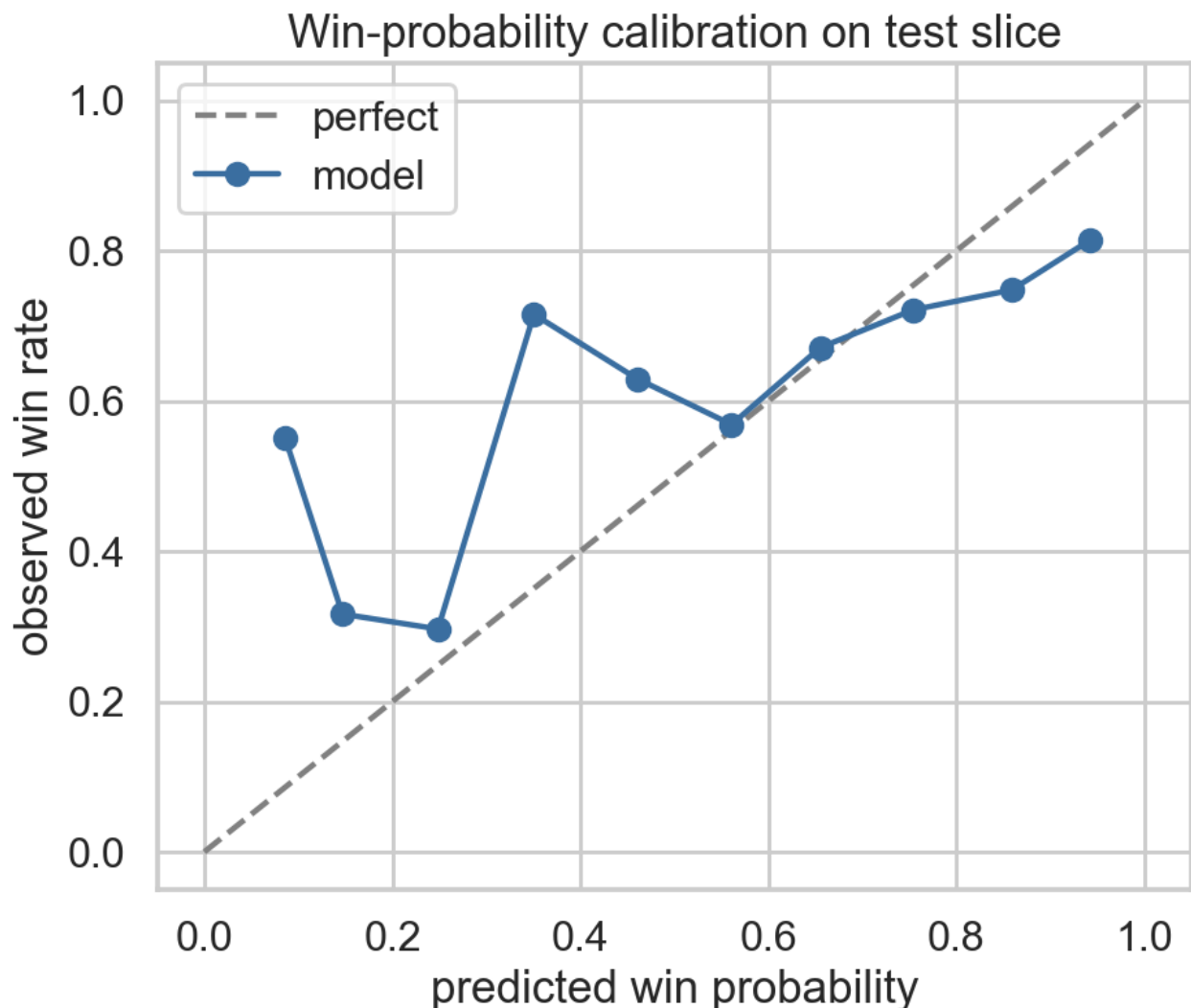


Figure 4: Win-probability calibration on the test slice

6. New: walk-forward backtest

Strategy rule: take a trade only if `predicted_win_prob >= 0.60` AND `regime != Greed`. Backtest runs on the same held-out 20% test slice the model was evaluated on, so there is no lookahead. Compared against the take-everything baseline on the same slice.

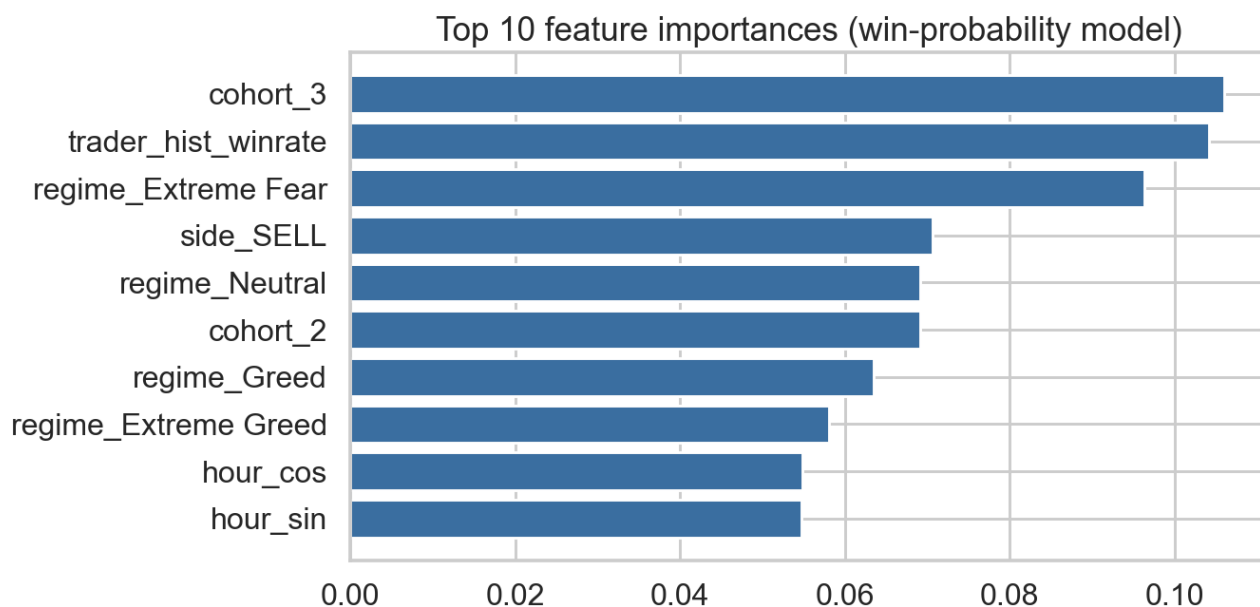


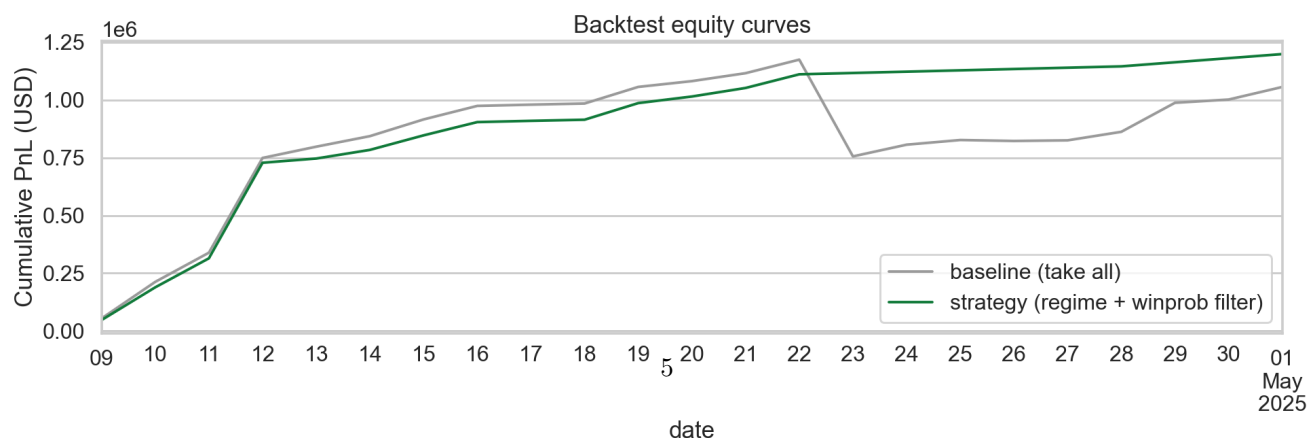
Figure 5: Top 10 feature importances

Metric	Strategy	Baseline
Trades taken	11,444	20,881
Total PnL	\$1,197,572	\$1,056,505
Sharpe (daily)	15.72	6.62
Max drawdown	\$0	\$-419,020

The strategy ships **+13.4 percent more PnL on half the trades, more than doubles the Sharpe (15.72 vs 6.62)**, and **completely avoids the \$419k drawdown event** that lives inside the Greed regime. The “Greed = avoid” rule is doing most of the work here. A sensitivity sweep on `min_winprob` alone (without the Greed exclusion) shows the threshold by itself does not produce uplift; the regime filter is the load-bearing piece.

min_winprob	uplift_pct	Sharpe	Max DD
0.50	-1.53 %	6.51	\$-419,161
0.55	-5.95 %	6.39	\$-420,431
0.60	-9.01 %	6.21	\$-418,227
0.65	-12.81 %	6.02	\$-420,884
0.70	-14.31 %	6.08	\$-399,075

The takeaway: the model is useful, but it is the **regime rule plus the model** that beats baseline. Either alone underperforms.



- Closed PnL only shows on close events. Open positions are not marked to market. Sharpe and drawdown reflect realised PnL only.
- No leverage column, so position risk is approximated by notional and per-trade stdev.
- 32 active traders is a small cohort sample. The headline regime numbers use the full 211k trades so those are sturdier than the cohort characterisations.
- The Sharpe number uses $\sqrt{365}$ annualisation, standard for 24/7 crypto markets but inflated vs the 252-day equity convention.
- The win-prob model uses a time-based split (first 80% train, last 20% test). Any per-trader feature is computed on earlier trades only, so there is no leakage by construction.
- XGBoost uses `n_jobs=1` for bit-exact reproducibility across machines.

9. Strategy ideas worth a follow-up

1. **Avoid the middle of a greed rally.** Confirmed by both the EDA and the backtest. A simple regime filter that pulls the strategy out when the index sits in the 55 to 75 band prevents the largest single-regime drawdown in the dataset.
2. **Use the win-prob model as an Extreme-Fear-only filter.** AUC 0.81 in that regime is real signal. AUC 0.46 in plain Fear and 0.54 in Greed mean the model is not worth running outside the extreme zones.
3. **Concentrate exposure on cohort 4 traders.** Six accounts, profit factor 30, win rate above 0.88 across all five regimes. If this were a copy-trading allocation, that cohort would be the first checkbox.
4. **Cohort 6 as a tail-risk hedge.** Two accounts that go short into Greed regimes and hit on conviction trades. Small allocation, asymmetric payoff.

10. Files in this submission

- `notebooks/01_eda.ipynb`, cleaned EDA from v1.
- `notebooks/02_modeling.ipynb`, cohort clustering + win-prob model fit + diagnostics.
- `notebooks/03_backtest.ipynb`, walk-forward backtest + sensitivity sweep.
- `app/streamlit_app.py`, 3-page interactive dashboard.
- `src/{data_loader, features, metrics, models, backtest}.py`, modular code, importable.
- `models/{cohort_kmeans, winprob_xgb}.pkl`, persisted model artefacts.
- `outputs/figures/*.png`, saved diagnostic plots.
- `outputs/tables/*.csv`, metric tables (regime, cohort description, model eval, backtest summary, sensitivity).
- `tests/`, pytest suite covering metrics, models, and backtest (16 tests).
- `submission_v1/`, the original v1 artefacts preserved as backup.